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Ipt 2 - image processing in radiography

Radiography and imaging sciences (Kenya Medical Training College)

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The x-ray exposure necessary to produce a radiograph of satisfactory diagnostic quality, commensurate with minimum exposure to the patient, depends not only on the exposure technique and film-screen combination employed but also on the handling and processing of the film. These require a good darkroom and proper developing techniques.

THE DARKROOM

Most modern x-ray departments use automatic film processors for film development. Nevertheless, good darkrooms are still an essential requirement whether they are used for manual processing of films or for loading automatic film processors. While specific details may vary from installation to installation, all darkrooms should include certain basic features:

1. DARKROOMS contain film processors, a loading bench, a film bin for storage of unexposed radiographic film, a film ID printer, safelights, and a pass box in the wall which allows transfer of film/cassettes to and from the darkroom, while ensuring no light is admitted to the darkroom.
2. The room *must* be completely light-tight.
3. If adjacent to a radiography room, the darkroom *must* be adequately shielded to ensure that exposure of personnel and film to x-radiation does not occur.
4. The darkroom *should* be designed to incorporate a lockable door, double doors, or a blackened maze entrance to ensure light tightness when undeveloped films are being handled.
5. A warning light *should* be located outside the darkroom, at the entrance, to indicate when the room is in use.
6. Safelights fitted with bulbs or correct intensity (7-15 WATTS) above the work areas, within the darkroom. The safelights *must* have filters appropriate to the specifications of the film used and *must* be positioned at the proper distances from work areas.
7. SAFELIGHTS Should be located at least 3 feet from the counters to prevent safelight fogging of exposed, unprocessed film.
8. Walls of the darkroom are painted a light color to help reflect the safelights.
9. Ample ventilation is required because of the presence of chemical fumes.

SAFELIGHTS

Kodak 6B filters are dark orange in color and are used for blue-sensitive film. This film is no longer in use. GBX filters are a deep cherry red color and are safe to use with green sensitive film. Safelights are only safe for a limited amount of time, and must contain the correct wattage bulbs, filters, and be located at least 3 feet from the worktable to not fog the undeveloped film.

FILM HANDLING

Films must be handled with clean, dry hands, and touched only at the corners. Dirt or chemical residue on hands can cause unwanted marks on the film or may stain the intensifying screens on the cassettes.

Mishandling of the unprocessed film can cause unwanted marks called **ARTIFACTS.** Film artifacts may be caused by cassette damage, dirty screens, careless handling, improper storage, or problems within the automatic processor. A sharp bending of the film can cause crescent or crinkle marks. Common film handling artifacts include creases, finger marks and scratches. If the humidity in the darkroom is too low static electricity can occur causing a crown static or tree static marks on the films. If films are improperly stacked, smudge static will occur. Cassettes should be reloaded as soon as possible after unloading, to prevent dirt

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from getting into the cassette. Cassettes that have been exposed to radiation must be kept separate from cassettes loaded with unexposed film to avoid the possibility of accidentally reusing an exposed cassette creating a double exposure (which will also result in increasing the radiation dose to the patient by having to repeat the image).

FILM PROCESING

Automatic film processors are very simple to use. The film is unloaded from the cassette and placed on the feed tray. The guide shoes then pull the film into the processor though a system of rollers to begin the development. Once the entire film has passed beyond the intake rollers, and so no longer visible on the feed tray, a tone or red light (or both) will signal that it is safe to feed the next film into the processor, or it is safe to turn on the lights. Feeding another film before the signal will result in an overlap of films, possibly ruining the quality of processing for both films, and may also cause a jam within the processor. Turning on the white light (or opening the door) before the signal may cause the trailing edge of film on the feed tray to become fogged. The finished film will drop into the receiving bin from 90 second to 3 minutes from the time it entered the processor.

The purpose of film processing is to change the silver halide crystals contained in the film emulsion (once exposed to x-ray or light photons) to black metallic silver. The LATENT image is then developed into a MANIFEST image.

Improper or careless processing of exposed radiographic films can cause films of poor diagnostic achieve full development of a film which has been exposed, using correct radiographic technique factors, the film must be processed in chemically fresh developer, at proper temperature and for sufficient time to ensure that the silver in the exposed silver halide crystals, in the film emulsion is completely reduced. If this is not done, the blackening of the film will not be optimum and the tendency will be to increase radiation exposure to achieve proper image density. To ensure proper processing of films, certain basic recommendations *should* be followed:

1. Manufacturer s recommendations with respect to strength of chemical solutions, temperature and time *must* be followed to ensure optimum development.
2. Developing solutions *should* be replenished as necessary and *should* be changed regularly, as required.
3. Developing solutions *should* be monitored regularly. Even unused developer deteriorates with time. Developer *should* not be used when I becomes necessary to develop significantly longer than the recommended times in order to obtain optimum film density.

DARKROOM FOG:

May be caused from unsafe safelights, white light leaks, excessive developer time, strength or temperature which will result in a decrease of radiographic contrast, and an unwanted increase in radiographic density.

FILM STORAGE

Unexposed radiographic films *must* be stored in such a manner that they are shielded from stray radiation (or light). Storage *should* be provided such that no film receives more than 0.2 milliroentgen of stray radiation before use. Films *should* be stored on end in a cool, dry area.

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CHEMICAL COMPONENTS OF PROCESSING SOLUTIONS

Developer Solutions

Reducing Agent:

Hydroquinone Reduce EXPOSED silver halide to black metallic silver

Metol or Elon

Phenidone

Activator:

Sodium Carbonate Softens gelatin, maintains alkaline pH (increase pH)

Hardener

GlutaraldehydePrevents over swelling of gelatin in automatic developer

Preservative

Sodium SulfiteAntioxidant ➡ prevents oxidation of developer

Restrainer

Potassium BromidePrevents chemical fog in new developer

Solvent

Water

Fixer Solution

Clearing Agent:

Ammonium ThiosulfateDissolves undeveloped silver halides

Tanning Agent:

Aluminum Salt Shrinks, hardens, preserves emulsion

Activator:

Acetic Acid Neutralizes developer, maintains acid pH (decrease pH)

Preservative:

Sodium Sulfite Prevents oxidation, prolongs solution life

Solvent:

Water

*Diagram of film path through automatic processor:

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AUTOMATIC FILM PROCESSING

Film emulsion consists of silver bromide (halide) crystals suspended in gelatin. The development process transforms the latent image to a ***manifest*** (visible) black metallic silver ***image.***

Automatic film processing is carried out by a machine that transports the x-ray film through the necessary chemical solution, at the same time providing agitation, temperature regulation, and chemical replenishment. Within the processor are the developer, fixer, and wash tanks, followed by the dyer.

Rapid processing is accomplished by the use of increased solution temperatures, which requires that a hardener be added to the developer to control excessive emulsion swelling.

Each of the processor systems accomplishes specific functions; a basic understanding of these systems is required so that the processor can be used correctly and efficiently. A properly maintained and monitored processor will ensure consistent radiographs that will retain their quality images over a long period of time (archival quality).

A. CHEMISTRY OVERVIEW

1. **Developer** ➪ The developer functions to convert the lantent (invisible) image into the manifest (visible) silver image by reducing the exposed silver bromide crystals to black metallic silver. Important factors affecting the development process are time (length of development), temperature (of the developer solution), and solution activity (strength, concentration).

The developer solution has an alkaline nature for optimal function of the reducing agents. Sodium or potassium carbonate provides the necessary alkalinity and serves as an activator (or accelerator) by swelling the gelatin emulsion so that the reducing agents are better able to penetrate the emulsion and reach the exposed silver bromide crystals.

The reducing agents are hydroquinone, which works slowly to build up blacks in the film areas of greater exposure, and phenidone, which quickly produces the gray tones in areas of lesser exposure. With respect to sensitometry, hydroquinone controls the shoulder (Dmax) of the characteristic curve, and phenidone controls the toe (Dmin) area.

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The developer solution, particularly the hydroquinone, is especially sensitive to oxygen. If the developer oxidizes, it becomes weaker and less effective. The preservative, sodium sulfite, or cycon, is added to the developer to prevent its rapid oxidation. A hood always covers the developer tank and replenisher solution. The solvent for the concentrated chemicals is water, used to dilute the concentrate to the proper strength.

Rapid processing is achieved through the use of high temperatures that accelerate the development process; however, high temperatures can cause excessive emulsion swelling. Since excessive swelling can result in roller transportation problems a hardener, traditionally Glutaraldehyde, is added to the developer to control the amount of emulsion swelling.

A restrainer, or antifog agent, is added to the developer to limit its activity to only the exposed silver crystals. The typical restrainer is potassium bromide. Without the restrainer, the developing agents would attack the unexposed crystals, creating chemical fog. Potassium bromide is frequently referred to as starter solution because it is added only to fresh, new developer. As films are developed, bromine ions are released from the emulsion into solution; thus, potassium bromide is not found in replenisher solution.

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2. **Fixer** ➲ The function of the fixer (hypo) is to clear the film of the unexposed, undeveloped silver bromide crystals. This process serves to protect the film from further exposure. The fixing or clearing agent is ammonium Thiosulfate.

The fixer is an acidic solution that functions to neutralize any residual developer carried over and provide the required acid medium for the hardener. Acetic acid provides the required acidic medium.

The fixer contains a hardener whose function it is to shrink and re-harden the gelatin emulsion, thus protectin it from abrasion and promoting archival quality. The most commonly used hardeners are potassium alum or aluminum chloride. Fixer preservative sis the same as that found in the developer, ie, sodium sulfite.

3. **Wash** ➲ The function of the wash is to ride the film of residual chemicals. Should chemicals remain in the emulsion (eg, as a result of defective wash cycle), the film will discolor with age. Since radiographic records are kept for a number of years, it is important that they have sufficient archival quality.

Cold water processors are, in general, less efficient in removing chemicals than warm water processors. Agitation during the wash process and large quantities of water help to rid the emulsion of chemical residue.

Summary

- Developer solution reduces the exposed silver bromide crystals to black metallic silver.
- The development process is greatly affected by development time and solution temperature and activity.
- Sodium or potassium carbonate provides the necessary alkalinity and functions as the solution activator by swelling the gelatin emulsion.
- Reducing agents are phenidone and hydroquinone.
- Sodium sulfite or cycon preserves the developer solution from excessive oxidation.
- Glutaraldehyde is a hardener, added to developer solution to control excessive swelling.
- Potassium bromide serves as an antifog agent and restrains the developer from attacking the unexposed silver bromide crystals.
- Potassium bromide is starter solution, and is not required in the replenisher solution.
- The fixing or clearing agent (ammonium thiosulfate) removes unexposed silver bromide crystals from the emulsion, preventing further exposure.
- Acetic acid provides the necessary acid medium for the fixer solution.
- Potassium alum or aluminum chloride serves to harden the film emulsion.
- The fixer preservative is sodium sulfite.
- Adequate washing of residual chemicals from the film emulsion is essential for good archival quality.

B. TRANSPORT SYSTEM

The transport system functions to convey the film through the different processor sections by means of a series of rollers driven by gears, chains, and sprockets. This is accomplished without damage to the film and at a prescribed speed, which determines the length of time films spends in each solution. The roller system also provides constant, vigorous agitation of the solution at the film surface. The entire conveyance system consists of the feed tray, crossover rollers, deep racks, turnaround assemblies, and receiving bin.

Film is aligned against one side of the feed tray as it is introduced into the processor. A sensor initiates solution replenishment as the film enters, and replenishment continues as the length of the film passes the sensor. Films should be fed into the processor along their short edge; feeding the film in the long way leads to over-replenishment and increased radiographic density.

Crossover racks are out of solution and bridge the gaps between developer and fixer, fixer and wash, and wash and dry sections of the processor. Crossover rollers must be kept free of crystallized solution that can cause film artifacts as the soft emulsion passes by. The last set of rollers in each solution section has a squeegee action on film emulsion, thus removing excess solution before film enters the next tank.

When the processor is not in use for a period of time, it is advisable to leave the lid open so that moisture can escape. Since the crossover rollers are out of solution, chemicals carried onto them by film can crystallize and should be cleaned off before the processor is used again.

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Turnaround assemblies are located at the bottom of the deep racks and serve to change the film direction as it changes from downward to upward motion. Guide shoes, or deflector plates, are also located where film must change direction. They will occasionally scratch film, leaving characteristic guide shoe marks, when they require adjustment.

When returning rollers to the processor after cleaning, care must be taken to seat them securely in their proper position. Transport problems (processor jam-up) will result if racks are misaligned.

C. REPLENISHMENT SYSTEM

As films travel from one processor solution section to another, chemical solution is carried away in the swollen film emulsion. It is the function of the processor replenishment system to keep solution tanks full. If solution level is allowed to lower, film-immersion time decreases and radiographic density and contrast changes will occur. Transport problems can also arise from inadequate replenishment, ie, if insufficient developer replenisher, the inadequate addition of hardener will result in excessive emulsion swelling. The essentially thicker film has difficulty transporting between the closely distanced rollers.

As film travels through the fixer, it accumulates residual developer solution; fixer solution also accumulates unexposed silver cleared from the emulsion. Wash water accumulates fixer. In these ways, the activity of each solution is depleted through continual use. Diminished solution activity can have the same effects as low solution levels. The replenishment system assures that proper solution concentration is maintained.

D. TEMPERATURE REGULATION

The temperature-regulation system functions to control the temperature of each section of the automatic processor. Developer is the most important solution temperature to regulate; in a 90-sec processor, developer temperature is usually maintained at 92º to 95º F. Once the correct developer temperature is established, it must be constantly maintained. Even a minor fluctuation (ie, 0.5º) in developer temperature can cause a visible change in radiographic density and contrast.

Developer temperature is thermostatically controlled and developer solution is circulated through a heat exchanger under the fixer tank. Thus, the fixer temperature is regulated (in cold water processors) by heat conducted from the developer solution. In older processors having stainless steel tanks, fixer temperature is regulated by heat convection from the neighboring developer solution.

E. RECIRCULATION SYSTEM

As replenishment chemicals are added to solution, the recirculation system provides agitation necessary for uniform solution concentration.

F. WASH AND DRYER SYSTEMS

Thorough removal of chemical solutions from the film emulsion is required for good archival film quality and is provided by the wash section of the automatic processor. Agitation of the water makes the process more efficient. Any residual chemicals will eventually result in film stain. Residual fixer will eventually stain the film a yellowish brown that ultimately obscures the image and diminishes the archival quality. Films can be tested (usually by the film manufacturer or distributor) to determine their degree of fixer retention.

The dryer section functions to remove water from the film by blowing warm, dry air over the film surface. Dryer temperature is usually 120º to 130ºF, sufficient to shrink and dry the emulsion without being excessive. Excessive heat and overdrying can cause film damage. If films emerging from a properly heated dryer are damp, the problem may be excessive emulsion swelling and water retention as a result of inadequate developer or fixer replenisher (hardener).

G. SILVER RECOVERY

X-ray film is expensive and represents a large part of a radiology department annual budget. About half of the film's silver remains in the emulsion after exposure and processing. The other half (unexposed silver)

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is removed from the film during the fixing process and most of it is recoverable through silver recovery methods. A drain is connected to the fixer tank and fixer is allowed to flow directly into a silver recovery unit or to a large centrally located receptacle.

Silver recovery is desirable for financial and ecological reasons. Fixer silver is toxic to the public water supply and environmental legislation makes persons responsible for its direct passage into sewer lines, or other means of improper disposal, subject to sever fines and penalties.

There are three types of silver recover methods. Used fixer enters a metallic displacement (or metallic replacement) cartridge and metallic silver is precipitated onto the steel wool within. This method of silver recovery is most useful for low volume locations.

Electrolytic silver recovery units (cells) pass an electric current through the fixer solution, causing silver to be plated onto a cathode cylinder of the unit. The silver is periodically removed by scraping it from the stainless steel cathode. Electrolytic cells are best used in locations having medium to high volume.

There are a number of chemicals that will precipitate metallic silver. In the presence of one of these chemicals (eg, sodium borohydride), metallic silver falls to the bottom of the tank and forms a sludge. This method of silver recovery is generally used only by large institutions having large, centralized receptacles or by professional silver dealers, who employ special techniques for separating the sludge or removing the entire tank.

FILM STORAGE CONSIDERATIONS

A. STORAGE CONDITIONS

The conditions under which x-ray film is stored can have considerable impact on the final radiographic image. The most common result of improper film storage is fog, which has a severely degrading effect on image quality.

1. **Temperature and Humidity.** Films should be stored at a temperature no greater than 70ºF. Temperature greater than 70ºF can accelerate the deterioration process and cause film fog. Atmospheric humidity should be kept between 40% and 60%. Excessively low humidity is conducive to the production of static electricity discharge. High humidity levels encourage the production of fog. An unopened (ie, hermetically sealed) bag of film protects the film from humidity but not excessive temperatures.
2. **Chemicals and Radiation.** Boxes of film must also be stored away from chemicals with fumes that can fog film emulsion. Film can be fogged if stored too close to radiographic rooms or radionuclides.
3. **Expiration Dates.** Each box of x-ray film is identified with an expiration date before which the film must be used in order to avoid age fog. When replenishing film supply, film boxes should be rotated so that the oldest film is used first.
4. **Position of Film Boxes.** Film boxes, should be stored in the upright position. If film boxes are stacked on one another, the sensitive emulsion (especially in the central portion) will be affected by pressure from the boxes above. Pressure marks (ie, areas of fog) are produced and result in loss of contrast in that area of the radiographic image. Larger size film boxes are particularly susceptible to this problem.
5. **Film Bin.** The film bin is a light-tight storage area where opened boxes of film are available for reloading empty cassettes. If a single door separates the darkroom from exterior white light, it is wise to have an automatic interlock system in place that prevents opening of the darkroom door while the film bin is open.

B. SAFELIGHT ILLUMINATION

Adequate and safe darkroom lighting is an essential part of ensuring quality radiographic images. A source of white light is required for cleaning and routine maintenance. The white light often has a safety devise to help prevent accidental film exposure. Safelight illumination must be appropriate for the type of film used and bright enough to provide adequate illumination and still not expose the sensitive emulsion (exposed film emulsion is about eight times more sensitive than unexposed emulsion).

1. **Types.** A frequently used safelight is the Kodak Wratten Series 6B, a brownish safelight filter, with a 7.5to 15-watt frosted light bulb placed 4 feet above the work surface. Another available safelight, which

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is somewhat brighter, is the Kodak GBX all purpose filter, which provides a more reddish illumination. This type of filter is often placed in the darkroom so that its light is directed upward toward the ceiling and reflected back down, thus reducing the chance of safelight fog.

2. **White Light Leaks.** Routine darkroom maintenance includes regular cleaning of all surfaces and walls and checks for white light leaks. When checking for light leaks, all darkroom lights must be turned off, adequate time given for eyes to adjust to the darkness, then a careful visual inspection made for white light leaks.

SUMMARY

- Film should be stored in a cool and dry environment, under 70ºF and between 40% and 60% humidity.
- Excessive temperatures cause film fog.
- Excessively low humidity encourages buildup of static electricity.
- Film should be stored away from radiation and chemicals.
- The film-box expiration date should be noted, and oldest film used first.
- Film boxes should be stored upright to avoid production of pressure marks.
- Kodak Wratten Series 6B and GBX darkroom filters are the most frequently used.
- Safelights should be placed 4 feet from the work surface with 7.5 to 15-watt light bulbs.

ARTIFACTS

Radiographically speaking, an artifact is a fault, blemish, or aberration in an x-ray image. It can be the result of improper handling, automatic processing, or use of defective radiographic accessories.

Cassettes, screens, and film must be handled carefully to avoid leaving fingerprints or production other film artifacts. Hands should be kept clean and dry, free from residue-leaving creams and powder from gloves. Film should be handled carefully by the corners when loading and unloading cassettes. The technologist should not slide film into or out of the cassette, as the friction can cause static electricity build-up. Cassettes should be numbered or otherwise identified so that artifact-causing problems can be located and removed.

IDENTIFICATION OF RADIOGRAPHS

Essential Information

Every radiograph must (for medicolegal reasons) include certain specific patient information.

- Patient's name or identification number
- The side marker, right or left
- The examination date
- Name of the institution

Other pertinent information may be included.

- Patient's age or birthdate
- Attending physician
- Time of day

When multiple films are taken of a patient on the same day, it is important that the time the radiographs were taken be included on the film. This permits the physician to chronologically follow the patient's progress.

1/20
B
The radiation-and light-sensitive layer of radiographic film is the _____ layer. (p. 171)
A. base
B. emulsion
C. supercoat
D. anticurl/antihalation

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2/20
C
Crossover is a radiographic problem because it decreases: (p 178)
A. contrast
B. density
C. recorded detail
D. film speed

3/20
A
Spectral sensitivity refers to the color sensitivity of: (p 176)
A. radiographic film
B. safe light filters
C. intensifying screens
D. safe light filters and intensifying screens

4/20
B
According to the Gurney-Mott theory of latent image formation, electrons liberated by radiation or light exposure are attracted to the: (p 173-174)
A. bromide
B. sensitivity speck
C. bound silver ions
D. gelatin

5/20
C
Silver halide crystals are found in the film's: (p 177)
A. base
B. supercoat
C. emulsion
D. anticurl/antihalation layer

6/20
C
Which of the following is the most common type of film for general radiographic examinations? (p 170 & 178)
A. direct exposure
B. screen, single-emulsion
C. screen, double-emulsion
D. rare earth

7/20

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C
Poor film-screen contact results in a loss of (p 191)
A. density
B. contrast
C. recorded detail
D. speed

8/20
D
The purpose of intensifying screens is to: (p 187)
A. increase radiographic density
B. increase recorded detail
C. decrease recorded detail
D. decrease patient dose
9/20
B
The most common phosphor material used in today's intensifying screens is/are: (p 186)
A. calcium tungstate
B. rare earth elements
C. silver halide
D. barium sulfate

10/20
D
The speed of an intensifying screen can be reduced by adding: (p188)
A. more phosphor
B. a reflecting layer
C. larger phosphor crystals
D. dye to the phosphor layer

11/20
B
What is the intensification factor for screens that require 5 mAs to produce the same density as produced by direct exposure using 150 mAs? (p 184 IF=exp req without screens/exp req with screens)
A. 5
B. 30
C. 50
D. 300
12/20
C
If 25 mAs is used with a 500 speed film-screen system to produce an optimal image, how much mAs is needed to produce the same density with a 100 speed system? (p 185 ma1/mas2 = relative screen speed2/relative screen speed1)
A. 5 mAs
B. 100 mAs
C. 125 mAs
D. 300 mAs

13/20

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A

Typically, as screen speed decreases, _____ decreases:(p184)

A. density

B. recorded detail

C. patient dose

D. x-ray exposure

14/20

C

Which of the following strategies reduces quantum mottle? (p190)

A. using a faster speed film-screen system

B. increasing kVp and reducing mAs

C. Reducing kVp and increasing mAs

D. Reducing kVp and reducing mAs

15/20

D

The wire mesh test tool is used to evalute: (p191)

A. screen speed

B. screen resolution

C. screen cleanliness

D. film-screen contact

16/20

D

The image receptor for computed radiography (CR) is composed of: (p193)

A. amorphous selenium

B. amorphous silicon

C. silver bromide

D. barium fluorohalide

17/20

A

Both types of DR flat panel detector digital imaging systems convert x-ray energy into: (p196)

A. electric charges

B. black metallic silver

C. chemical energy

D. light energy

18/20

D

The latent image that consists of electrons trapped in the phosphor layer is found with: (p197)

A. direct conversion DR digital systems

B. indirect conversion DR digital systems

C. film-screen radiography

D. CR

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19/20
A
Exposed CR imaging plates should be processed within _____ hour(s) to avoid any loss of image quality because of fading: (p195)
A. 1
B. 5
C. 24
D. 48

20/20
C
Thin film transistors (TFTs) are used in DR systems to: (p195)
A. produce a flash of light
B. convert the x-ray energy to an electric charge
C. temporarily store the electric charge
D. trap the electrons in the photostimulable phosphor layer.

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Rad 141Test 1
Rad 141 Principles of Radiographic
Question Answer
The principle characteristic of the emulsion that makes it particularly x-ray sensitive is its: Sensitivity
The radiographic film base appears blue So that image viewing is more comfortable
The emulsion that is commonly used in x-ray film consist of which of the following Silver halide and gelatin
A common base of contemporary x-ray film Polyester
Which of the following are the two basic parts of radiographic film Base and emulsion
Dimensional stability is the property of maintaining the size and shape of which of the following Base
The principle purpose of gelatin in the emulsion is to: Support the silver halide crystals uniformly
X-ray film is sensitive to which of the following Only x-rays, gamma rays, and, visible light
Which of the following is a principle component of the emulsion Silver bromide
The latent image actually is formed in the Silver halide crystal
During the photographic process, metallic silver accumulates T the: Sensitivity center
The latent image at the crystal level: Is a collection of silver atoms
The term latent image actually refers to which of the following An unprocessed image
The sensitivity center is usually silver..... Sulfide
When the latent image is processed, what type of image does it become Excited
What do we call the radiation exciting the patient that is responsible for latent image formation Image-forming, remnant
The interstitial atoms in a crystal of silver halide include which of the following Bromide, silver, iodine
If one observes a film immediately after exposure, what is seen A latent image
Which of the following theories best explains the photographic effect Gurney-Mott
The film base that provides the increased strength and thinner form needed for automatic processing is: Polyester
The emulsion of a radiographic film is composed of silver halide and... Gelatin
The atoms comprising the silver halide crystal are bond together by what kind of bond Ionic bonds
How should radiographic film be stored, in order to prevent pressure artifacts on the film On-end
Opened boxes of film should be stored at what temperature, and between what percent of of relative humidity <68°F 20°C and 40-60% humidity

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Characteristics of resolution Greater the 1p/mm = Greater resolution
Characteristic of speed Greater resolution = lesser speed = Greater exposure Lesser resolution = Greater speed = Lesser exposure
Characteristic of contrast Greater subject contrast = lesser contrast film Lesser subject contrast = greater contrast film
Characteristic of latitude Greater contrast film = narrow latitude Lesser contrast film = wide latitude
Radiographic intensifying screen speed increases with increasing... Phosphor thickness
Direct-exposure, onscreen technique results in better... Spatial resolution
The resolution achievable with slow speed radiographic intensifying screen is approximately 15lp/mm
Which of the following factors controlled by the radiologic technologist affects radiographic screen speed the most kVp
The intensifying factor characteristic Greater for rare earth screens than for calcium tungstate screens
A phosphor currently used in radiographic intensifying screens is Gadolinium oxybromide
Radiographic intensifying screens are used with radiographic film principally to Reduce pt exposure
Because x-ray interaction with a phosphor produces light isotropically The light spreads out
Mammography conducted with detail screens result in entrance skin exposures (ESE) as low as 300mR (3mGya). The intensification factor for these screens is about 30. If such an examination were conducted as a onscreen procedure, what would be the ESE? 9000mR
Which of the following is a rare earth element Gadolinium, yittrium, lanthanum
The principle advantage of rare earth screens over calcium tungstate screens is Faster speed
Which of the following element is a green light emitter Gradolinium
Which of the following principally contributes to the increased speed of rare earth screens Conversion efficiency
The routine care of radiographic intensifying screens requires that they be Periodically cleaned
Phosphor afterglow Aka Lag
The pt exposure for a Direct-exposure CXR is 210mR. If a par-speed screen are used, the exposure required is 15mR. The intensification factor is: 14
A Direct-exposure radiograph requires 650mR. If the intensification factor of rare earth screens was 200, whT would be the patient exposure if such screens were used 3.25mR
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3/263
ascites
A condition that leads to an abnormal collection of fluid in the abdominal cavity is:
Select one:
a. Ascites
b. Peritonitis
c. Pleuritis
d. Ileus

4/263
1, 2, and 4
A CR reader utilizes which of the following in producing the digital image signal?
1. photomultiplier tube
2. laser beam
3. charged coupled device

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4. light channeling guide
Select one:
a. 1, 2, and 3
b. 2, 3, and 4
c. 1, 2, and 4
d. 1, 2, 3, and 4

5/263
ketoacidosis
A diabetic condition with incomplete fat metabolism due to too little insulin and evidenced by mental confusion or diabetic coma is:
Select one:
a. ketoacidosis
b. hypoglycemia
c. diabetic shock
d. insulin shock

7/263
quantum mottle
A loss in image quality sometimes produced when a very fast speed intensifying screen is used and a very low number of x-ray photons creates the image is called:
Select one:
a. high resolution
b. quantum mottle
c. intensification factor
d. exposure index failure

5/263
hypovolemic
A type of shock due to low blood volume caused by conditions such as trauma, burns, GI or surgical bleeding and evidenced by weak and rapid pulse rate, cool or clammy skin and increased respirations is called:
Select one:
a. anaphylactic
b. hypovolemic
c. cardiogenic
d. septic

22/263
Setting up a magnetic attraction that rotates the anode disk
An induction motor and stator serve the x-ray tube by:
Select one:
a. Producing free electrons
b. Focusing the electron cloud or stream onto the target
c. Opening and closing the window
d. Setting up a magnetic attraction that rotates the anode disk

23/263
carina
Another name for the trachea's point of bifurcation is the:
Select one:
a. Carina

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b. Cricoid
c. Epiglottis
d. Thyroid

33/263
blunt trauma to the chest
Cardiac tamponade would likely be caused by:
Select one:
a. blunt trauma to the chest
b. reaction to certain medications
c. radiation sequela
d. pulmonary edema

34/263
Classical scattering has no change in wavelenght (energy of the photon).
Classical scattering is unlike Compton scattering in that:
Select one:
a. Classical scattering has no change in x-ray direction.
b. Compton scattering has no change in x-ray direction.
c. Classical scattering has no change in wavelenght (energy of the photon).
d. Compton scattering has no change in wavelenght (energy of the photon).

5/263
creasing or bending the film prior to processing
Crescent shaped black artifacts on a conventionally wet processed radiograph may be commonly caused by:
Select one:
a. static electricity
b. creasing or bending the film prior to processing
c. low humidity levels in the darkroom
d. improper development

39/263
pulmonary edema
For a patient with chronic obstructive pulmonary disease (COPD), allowing the IV infusion to run too fast could result in:
Select one:
a. pneumothorax
b. ischemic stroke
c. Legg-Perthes disease
d. pulmonary edema

40/263
improperly erased PSP CR plate
"Ghosting" of the digital image is most often the result of:
Select one:
a. high SNR and low DQE
b. use of an incorrect LUT

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c. improperly erased PSP CR plate
d. manual processing of an image whose signals fall outside the acceptable exposure index range

42/263
From the prone position, rotate the hand radial side up.
How should the hand be positioned for the PA medial oblique projection of the hand?
Select one:
a. From the prone position, rotate the hand ulnar side up.
b. From the prone position, rotate the hand radial side up.
c. From the supine position, rotate the hand ulnar side up.
d. From the supine position, rotate the hand radial side up.

47/263
Barium enema
If a patient was scheduled for an upper gastrointestinal series and a barium enema on the same day, which exam should be performed first?
Select one:
a. Barium enema
b. UGI
c. The order of the exams does not matter.
d. Both exams cannot be performed on the same day.

48/263
turn the grid lengthwise
If during a trauma projection you are using a grid and need to cross-angle the CR, you will need to:
Select one:
a. Turn the grid lengthwise.
b. Turn the grid crosswise.
c. Angle the grid diagonally.
d. It does not matter much.

, 1,2 and 3
In addition to using a grid, other techniques or devices that will increase radiographic contrast include:
1. collimator
2. air gap technique
3. compensating filter
Select one:
a. 1 and 2 only
b. 2 and 3 only
c. 1 and 3 only
d. 1, 2, and 3

54/263
5 degrees caudad

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In an attempt to remove shoulders on the cross-table (C/T) lateral C-spine, the radiographer would perform a swimmer's view. If the patient was unable to offset the shoulders, the radiographer could adjust the CR by angling:
Select one:
a. 5 degrees caudad
b. 5 degrees cephalic
c. Perpendicular
d. 15-20 degrees caudal

56/263
decrease
In comparing film screen images with digital images produced with indirect digital (cassette based) radiography or CR, there is generally a(an) _____ in recorded detail with film screen images as compared to digital images.
Select one:
a. increase
b. decrease
c. equal result

57/263
silver bromide
In order to create a latent image on the film, which of the following compounds becomes disrupted during the photographic effect?
Select one:
a. Sodium sulfite
b. Silver bromide
c. Chromium alum
d. Calcium tungstate

59/263
occipital bone
In the AP axial position of the skull, the central ray is directed 30 degrees caudal to the orbitomeatal line (OML) and passes between the external auditory meatuses, which of the following anatomical structures is best demonstrated:
Select one:
a. Frontal bone
b. Facial bones
c. Temporal bone
d. Occipital bone

66/263
central nervous system
In total body irradiation, the radiation syndrome which results most often in death due to radiation doses above 50 Gy is the _____ syndrome.
Select one:

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a. hematologic
b. bone marrow
c. gastrointestinal
d. central nervous system

67/263
Plantodorsal axial projection of the os calcis
In which of the following positions/projections will the talocalcaneal joint be visualized?
Select one:
a. Dorsoplantar projection of the foot
b. Plantodorsal axial projection of the os calcis
c. Medial oblique position of the foot
d. Lateral foot

68/263
emphysema
In which of the following pathologic conditions should the technologist decrease normal exposure factors to insure appropriate image density?
Select one:
a. pulmonary edema
b. Ascites
c. pleural effusion
d. emphysema

70/263
2 and 3 only
Medial oblique of the knee demonstrates:
1. Medial condyles and epicondyles
2. Fibular head and neck free of superimposition.
3. Lateral condyles and epicondyles
Select one:
a. 1 only
b. 2 only
c. 3 only
d. 1 and 2 only
e. 2 and 3 only

86/263
a shorter SID is used and larger field sizes are necessary
The anode heel effect has greatest application in improving image quality when:
Select one:
a. the part to be radiographed has uniform thickness and part density
b. smaller receptor sizes such a 8 x 10 inches are utilized
c. OID is more than 6 inches
d. a shorter SID is used and larger field sizes are necessary

93/263

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medial oblique elbow
The coronoid process should be visualized in profile in which of the following positions?
Select one:
a. Scapular Y
b. AP Scapula
c. Medial oblique elbow
d. Lateral oblique elbow

103/263
Effective focal spot
The focal spot size measured directly under the anode target and projected onto the patient and image receptor is referred to as the:
Select one:
a. Space charge
b. Effective focal spot
c. Actual focal spot
d. Anode heel

106/263
focused
The grid design that matches the divergence of the beam so that lead strips are scanted from perpendicular as they move from the center of the grid is the _____ grid.
Select one:
a. linear
b. cross hatch or criss cross
c. focused
d. parallel

107/263
grid focusing range
The ideal SID to prevent excessive grid cut-off in the use of a focused grid is referred to as:
Select one:
a. grid efficiency
b. grid frequency
c. grid focusing range
d. grid clean up

108/263
Spina bifida
The Kite method is done to demonstrate what congenital abnormality?
Select one:
a. Spina bifida
b. Maran's syndrome
c. Talipes equinavarous
d. Arthritis

109/263
Splenic flexure

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The LAO position of the large bowel is used to "open up" and best demonstrate which of the following?
Select one:
a. Splenic flexure
b. Hepatic flexure
c. Sigmoid colon
d. Ileocecal valve

110/263
Emulsion
The layer of radiographic film which converts x-ray or light photons into a latent image is the _____ .
Select one:
a. Base
b. Superlayer
c. Activator
d. Emulsion

112/263
T2-3 interspace
The manubrial notch is approximately at the same level as the:
Select one:
a. Fifth thoracic vertebra
b. T2-3 interspace
c. T4-5 interspace
d. Costal

118/263
Relationship of the pedicle and the lumbar body.
The obliquity used to demonstrate the zygapophyseal joints of the lumbar spine is 45 degrees. When evaluating the resultant image, how is the correctness of the obliquity determined?
Select one:
a. Relationship of the pedicle and the lumbar body.
b. Space between adjacent vertebral bodies.
c. Distance between the ear and tip of the front leg of the "Scottie Dog".
d. Amount of curve in the lumbar spine.

119/263
RPO
The patient position used to best demonstrate the splenic flexure is:
Select one:
a. LPO
b. PA
c. RPO
d. RAO

122/263
Toe
The portion of the characteristic curve on the graph that represents the extremely small exposures that result from an exam or sensitometric exposure to the film is called the _____ .
Select one:
a. Toe

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b. Shoulder
c. Base plus fog
d. Straight line portion

124/263
spatial distortion
The principle of distortion which can be defined as the misrepresentation of the three-dimensional relationship of the structures within the body where overlying structures are shown as separated is called:
Select one:
a. image noise
b. spatial distortion
c. elongation
d. foreshortening

126/263
wire mesh
The quality assurance test tool which evaluates film screen contact is the:
Select one:
a. spinning top
b. pin hole camera
c. wire mesh
d. resolution grid

131/263
kVp
The technical factor that determines the potential difference or EMF between the cathode and anode thereby controlling the kinetic energy of the projectile electrons is:
Select one:
a. kVp
b. mA
c. Seconds
d. keV

140/263
Coned PA patella
To demonstrate a vertical patellar fracture, which one of the following would be best:
Select one:
a. Lateral patella
b. Coned PA patella
c. Coned AP patella
d. Medial PA oblique knee

142/263
Epiglottis
Tracheostomies are performed just below the ring of cartilage forming the inferior margin of the larynx and the area known as the:

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Select one:
a. Carina
b. Cricoid
c. Epiglottis
d. Thyroiod

144/263
Pneumothorax
What is the condition where free air enters the pleural cavity?
Select one:
a. Pleurisy
b. Pleural effusion
c. Pneumothorax
d. Pneumonia

145/263
Pleural effusion
What is the name of the condition where fluid enters the pleural cavity?
Select one:
a. Pleurisy
b. Pleural effusion
c. Pneumothorax
d. Pneumonia

147/263
Petrous pyramids are projected in the lower half of the maxillary sinus.
What is the primary evaluation criteria for the modified waters projection?
Select one:
a. Petrous pyramids fill the orbits.
b. Petrous pyramids fill the lower 1/3 of the orbits.
c. Petrous pyramids are projected in the lower half of the maxillary sinus.
d. Petrous pyramids are projected at the infraorbital rim.

149/263
Sphenoid effusion
What radiographic sign within the skull may indicate a basilar skull fracture as seen on the lateral projection?
Select one:
a. Maxillary effusion
b. Sphenoid effusion
c. Frontal effusion
d. Ethmoid effusion

150/263
Right lateral Decubitus
What specific lateral decubitus would be performed if a pneumothorax is suspected in the left lungs?
Select one:
a. Left lateral decubitus

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b. Right lateral Decubitus
c. Dorsal decubitus
d. Ventral decubitus

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151/263
Increases
When cell poliferation rate increases, cellular radiosensitivity:
Select one:
a. Increases
b. Remains unchanged
c. and metabolism usually decreases.
d. Decreases

155/263
Scaphoid, lunate, pisiform, and triquetrum.
Which bones are located in the proxima row of the wrist?
Select one:
a. Scaphoid, lunate, capitate, and hamate.
b. Scaphoid, lunate, pisiform, and triquetrum.
c. Trapezium, trapezoid, capitate, and hamate.
d. Trapezium, trapezoid, pisiform, and triquetrum.

156/263
Radial notch, olecranon process, and coronoid process
Which bony structures are located on the proximal end of the ulna?
Select one:
a. Radial notch, styloid process, and ulnar head.
b. Radial head, olecranon process, and ulnar head.
c. Radial head, styloid process, and coronoid process.
d. Radial notch, olecranon process, and coronoid process

157/263
Mitosis phase
Which cellular life cycle phase is the most sensitive to radiation exposure?
Select one:
a. G1 phase
b. G2 phase
c. S phase
d. Mitosis phase

158/263
silver halide crystals in gelatin
Which components make up the emulsion of x-ray film?
Select one:
a. silver halide crystals in the polymer matrix
b. gadolinium phosphor crystals in gelatin

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c. silver halide crystals in gelatin
d. gadolinium phosphor crystals in the polymer matrix

159/263
Left lateral decubitus
Which decubitus position best demonstraste free air in the abdominal cavity?
Select one:
a. Left lateral decubitus
b. Right lateral decubitus
c. Dorsal decubitus
d. Ventral decubitus

163/263
Carpometacarpal
Which joint is the most distal articulation of the wrist?
Select one:
a. Intercarpal
b. Radiocarpal
c. Carpometacarpal
d. Metacarpophalangeal

165/263
1 & 3 only
Which of the following are on the lateral side of the humerus?
1. Capitulum
2. Trochlea
3. Greater tubercle
Select one:
a. 1 & 2 only
b. 1 & 3 only
c. 2 & 3 only
d. 1, 2, & 3

166/263
1 & 3 only
Which of the following methods will demonstrate the intercondyloid fossa?
1. Beclere method
2. Settegast method
3. Camp-Coventry method
Select one:
a. 1 & 2 only
b. 1 & 3 only
c. 2 & 3 only
d. 1, 2, & 3

167/263
Protect the IR with plastic.
Which of the following procedures should the radiographer perform if the trauma patient has bleeding wounds?
Select one:

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a. Protect the IR with plastic.
b. Enclose the IR inside a pillowcase.
c. Ask ED personnel to assist with procedure.
d. Cover the wounds with gauze and tape.

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168/263
Pneumothorax
Which of the following is considered a destructive pathology and would require a decrease in exposure?
Select one:
a. Congestive heart failure
b. Pneumonia
c. Pneumothorax
d. Pleural effussion

170/263
congentital abnormalities
Which of the following radiation responses in utero is most likely when exposure occurs during organogenesis?
Select one:
a. childhood cancer
b. congentital abnormalities
c. leukemia
d. neonatal death
e. prenatal death

173/263
double strand breaks
Which of the following types of radiation damage to DNA results more often with less efficient cellular repair, slower repair and the highest correlation with cell death?
Select one:
a. prodromal interaction
b. single strand breaks
c. ssublethal
d. double strand breaks

176/263
neurons
Which of the following are the most resistant cells to radiation damage?
Select one:
a. spermatogonia
b. neurons
c. lymphocytes
d. erythroblasts

183/263
AP or PA chest

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Which of the following radiologic exams would be most helpful in demonstrating the correct position of a swan-ganz catheter?
Select one:
a. AP and Lateral femur
b. KUB
c. AP or PA chest
d. ventral decubitus abdomen
185/263
Silver bromide
Which of the following silver halides is present in the largest numbers in film emulsion?
Select one:
a. Silver sulfite
b. Silver iodide
c. Silver sulfide
d. Silver bromide

195/263
The hand is supinated.
Which positioning maneuver should be done initially to properly position for the AP projection of the humerus?
Select one:
a. The hand is supinated.
b. The hand is pronated.
c. The hand is true lateral.
d. The elbow is flexed 90 degrees.

197/263
Lateral
Which projection best demonstrates the intervertebral foramina of the thoracic spine?
Select one:
a. AP
b. Anterior oblique
c. Lateral
d. Posterior oblique

198/263
Lateral
Which projection of the knee best opens the femoropatellar space?
Select one:
a. AP
b. External oblique
c. Internal oblique
d. Lateral

199/263
Head and tuberosity
Which structures are located on the proximal radius?
Select one:
a. Olecranon process and styloid process
b. Head and tuberosity
c. Head and styloid process

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d. Olecranon process and tuberosity

200/263
Off-focus radiation
X-rays created on locations in the tube anode other than the target are called:
Select one:
a. Remnant radiation
b. Off-focus radiation
c. Brems rays
d. Characteristic radiation

201/263
to open up the joint spaces
A 15 degree cephalic angle is used for an AP projection of the cervical spine. Why?
Select one:
a. To demonstrate the top three cervical vertebrae
b. To open up the joint spaces
c. To prevent superimposition of the
mandible
d. To demonstrate the vertebral arches

202/263
pisiform
A 45 degree semisupination oblique, anteroposterior projection of the wrist, would demonstrate the _____ free from superimposition.
Select one:
a. Navicular
b. Pisiform
c. Trapezium
d. Os magnum

203/263
Intestinal tract perforation
A barium sulfate suspension should not be used for a colon examination in a patient with suspected:
Select one:
a. Rectal bleeding
b. Intestinal tract perforation
c. Vomiting
d. Polyps

205/263
Orthostatic hypotension
A decrease in blood pressure which occurs on suddenly arising from the recumbent position is called:
Select one:
a. Orthostatic hypotension
b. Epistaxis
c. Vertigo
d. Syncope

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207/263
AP or PA and lateral
A foreign body within the soft tissues of an extremity requires which of the following projections>
Select one:
a. AP or PA and lateral
b. Both obliques
c. Tangential and lateral views
d. High kVp imaging

209/263
Quantum mottle or image noise
A loss in image quality sometimes produced with very fast intensifying screens when a very low number of x-ray photons creates the image is called _____ .
Select one:
a. Quantum mottle or image noise
b. Absorption deficiency
c. High conversion efficiency
d. High resolution

210/263
Crossover or parallax effect
A loss of image sharpness occurs when light from the screens passes through the film base to the other film emulsion and deposits a magnified and less sharp image. This occurence is called _____ .
Select one:
a. Crossover or parallax effect
b. Narrow exposure latitude
c. Capturing a grid
d. Grid cutoff

211/263
Right lateral decubitus chest
A patient entered the ER with possible right hemothorax. The patient is unable to stand or sit erect. Which of the following projections would best demonstrate this pathology.
Select one:
a. Apical lordotic
b. Ventral decubitus chest
c. Left lateral ducubitus chest
d. Right lateral decubitus chest

216/263
Xiphoid process
A radiologist has requested a spot radiograph of the tenth thoracic vertebra. The external landmark that would be most helpful for locating the level of the tenth thoracic vertebra is:
Select one:

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a. Costal margin
b. Xiphoid process
c. Iliac crests
d. Sternal angle

217/263
cystic duct, common bile duct
A stone which formed in the gallbladder would travel to the duodenum in what order?
Select one:
a. cystic duct, common bile duct
b. Left hepatic duct, common bile duct, cystic duct
c. Cystic duct, common hepatic duct, common bile duct
d. None of the above

220/263
AP erect, both shoulders
Acromioclavicular separation is best demonstrated in which of the following:
Select one:
a. AP recumbent, affected shoulder
b. AP recumbent, both shoulders
c. AP erect, affected shoulder
d. AP erect, both shoulders

234/263
Diet, cathartics, suppositories, or enemas
Common preparation for the examination of the intestinal tract may include:
Select one:
a. Diet, exercise, suppositories, or enemas
b. Diet, jogging, suppositories, or fasting
c. Diet, cathartics, drip I.V. injections
d. Diet, cathartics, suppositories, or enemas

237/263
Gastric or bowel mucosa
Double contrast examinations of the stomach or large bowel are performed to better visualize:
Select one:
a. Position of the organ
b. Size and shape of the organ
c. Diverticula
d. Gastric or bowel mucosa

238/263
Entrance roller assembly
During automatic processing, the activation of the developer replenishment pump microswitch is accomplished when the film comes in contact with the:
Select one:

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a. Crossover assembly between fixer and wash tanks
b. Guideshoes located between developer and fixer tanks
c. Entrance roller assembly
d. Developer squeegee assembly

246/263
medial
For a lateral radiograph of the fifth digit, which surface of the finger is in contact with the cassette?
Select one:
a. Palmar
b. Volar
c. Medial
d. Lateral

247/263
pulmonary edema
For a patient with chronic obstructive pulmonary disease (COPD), allowing an IV infusion to run too fast could result in:
Select one:
a. Pneumothorax
b. Pulmonary edema
c. Ischemic stroke
d. Legg-Perthes disease

between the patient and the image receptor
Radiographic grids must be placed
squeegee rollers
During automatic processing, solution carry over between the developer and fixer tanks is prevented by the
True
True or False
the use of intensifying screens will permit a shorter exposure time than without screens
3/40

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Terms in this set (40)
compensating filter
A device that is commonly used to equalize the various density differences within a patient is a
between the patient and the image receptor
Radiographic grids must be placed
squeegee rollers
During automatic processing, solution carry over between the developer and fixer tanks is prevented by the
True
True or False
the use of intensifying screens will permit a shorter exposure time than without screens
fluorescence
visible light emitted only during stimulation of a phosphor is called
true
True or False
Conversion efficiency contributes to the increased speed of rare earth screens

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True
True or False
differential absorbtion or the quantity of xrays absorbed by the subject is dependent upon, the atomic # of the tissue, kvp, and tissue density
after it is exposed to radiation
because darkroom safelights are not 100% effective in preventing light fog, it is important to know that a film is most sensitive to fog
Green
intensifying screens that contain rare earth phosphors emit what color light?
decreased # of xrays striking the film, random light conversion process
quantum mottle is caused by
100 mR/hr
The max amount of leakage radiation permitted from a radiographic tube housing may not exceed blank at a distance of one meter.
increases
As field size is increased, scatter radiation
silver halide crystals and gelatin
the 2 components of the radiographic film emusion are
A- Gadolinium
which of the following is a rare earth element:
A- gadolinium. B- Nobelium. C- Thalium. D-Thulium. E-Palladium.
Field Size
which of the following factors that affect scatter radiation can be controlled by the rad tech;
Added filtration, field size, inherent filtration, mAs, or patient thickness.

Terms in this set (18)
Which of the following is an assembly that consists of one master roller, several planetary rollers, and two guide shoes?
a.) A crossover rack
b.) A detector assembly
c.) A roller subassembly
d.) A transport rack
e.) A turnaround assembly
e.) A turnaround assembly
Replenishment of which tank in the processing system is most important?
a.) Developer
b.) Dryer
c.) Fixer
d.) Wash
e.) Wetter
a.) Developer
When a film is inserted into an automatic processor:
a.) A microswitch grips it.
b.) Guide shoes grip it.
c.) It should be centered on the feed tray.
d.) The long dimension should be against the rail.
e.) The short dimension should be against the rail.
e.) The short dimension should be against the rail.
Adequate drying is necessary to:
a.) Complete development

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b.) Complete fixation
c.) Obtain adequate contrast
d.) Reduce artifacts
e.) Strengthen the base
d.) Reduce artifacts
The fastest automatic processors can process a film in:
a.) 30 seconds
b.) 90 seconds
c.) 180 seconds
d.) 5 minutes
e.) 7 minutes
a.) 30 seconds
Between the fixing tank and the wash tank, the film passes through a:
a.) Crossover rack
b.) Drying chamber
c.) Receiving bin
d.) Transportation rack
e.) Turnaround rack
a.) Crossover rack
Which of the following is not one of the major systems of an automatic processor?
a.) The development system
b.) The drying system
c.) The electrical system
d.) The replenishment system
e.) The transport system
a.) The development system
The principal purpose of the circulation system is to:
a.) Agitate the chemistry
b.) Agitate the film
c.) Control chemistry concentration
d.) Control chemistry temperature
e.) Replenish the chemistry
a.) Agitate the chemistry
Which segment of the circulation system is most important to archival quality?
a.) Developing
b.) Drying
c.) Fixation
d.) Washing
e.) Wetting
c.) Fixation
Underreplenishment of the developer will result in:
a.) A decrease in contrast
b.) A reduction in spatial resolution
c.) An improvement in spatial resolution
d.) An increase in contrast
e.) No change in contrast
a.) A decrease in contrast
Replenishment tanks should have close-fitting floating lids primarily:
a.) For control of replenishment rate
b.) For control of temperature
c.) To control aerial oxidation

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d.) To easily monitor fluid level
e.) To reduce splash hazard
c.) To control aerial oxidation
Dry-to-drop time refers to:
a.) Development time
b.) Exposure-to-viewbox time
c.) Passbox-to-receiving bin time
d.) Time from feed tray to receiving bin
e.) Time in the passbox
d.) Time from feed tray to receiving bin
The segment of the circulation system that is most likely to contain a filter is the:
a.) Developer
b.) Dryer
c.) Fixer
d.) Wash cycle
e.) Wetting cycle
a.) Developer
If the power of the drive motor is transferred through a chain, the connecting device is usually a:
a.) Gear
b.) Pulley
c.) Roller
d.) Spring
e.) Sprocket
e.) Sprocket
Control of the replenishment system is accomplished by a:
a.) Drive motor
b.) Float valve
c.) Microswitch
d.) Timer
e.) Transport rack
c.) Microswitch

1/33
Recorded detail is defined as _____.
A) accuracy of structural lines recorded
B) visibility of the structural lines
C) misrepresentation of the shape of the structural lines
D) amount of structural lines
AHome Flashcards Preview
Rad Expo Final
The flashcards below were created by user joesaflea on FreezingBlue Flashcards.

Which 2 factors affect size distortion? (Select all that apply)

Select one or more:
a. OID
b. Focal spot size
c. Resolution
d. SID
e. Film screens

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f. Patient motion
g. SOD
OID
SID
Name 4 ways technologists can help control patient motion.
Four ways that technologists can help control patient motion are: using a short exposure time, communicate clearly with patient about breathing, use physical immobilization tools, and use others to help immobilize when needed.
Which 3 factors affect geometric unsharpness? (Select all that apply)

Select one or more:
a. Resolution
b. Patient motion
c. OID
d. SOD
e. Film screens
f. Focal spot size
g. SID
Focal spot size
OID
SID
Which is better at spatial resolution?

Select one:
a. film
b. digital
a. film
Based on the following factors, what is the geometric unsharpness? (In mm)
FSS=0.6mm, SID=62 inches, OID=2 inches
0.02 mm
A PA projection of the hand was produced with an SID of 40 inches and an OID of 0.5 inches. What is the MF? (with 3 decimal places)
1.013
Describe how poor film-screen contact will increase image unsharpness.
Light transmitted from the intensifying screen will diverge from its origin and so it is important to have good film-screen contact. Poor film-screen contact will expose parts of the film to light that shouldn't be exposed because of that divergent light beam and the space between film and screen.
A PA projection of the chest was produced with an SID of 72 inches and an OID of 4 inches. What is the MF? (with 3 decimal places)
1.059
To increase recorded detail OID should be maximized.

Select one:
True
False
False
Radiographic images of structures are always magnified when compared to the actual structure.

Select one:
True
False

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True
Why will radiographs always have some degree of unsharpness?
Radiographs will always have some sort degree of unsharpness because of all the different variables required to make an accurate radiograph including geometric unsharpness, receptor unsharpness, and motion unsharpness.
Size distortion is affected by:
1. SID
2. OID
3. Tube angulation

Select one:
A. 2 & 3 only
B. 1 & 3 only
C. 1, 2, & 3
D. 1 & 2 only
D. 1 & 2 only
Which factor has the most detrimental effect on recorded detail?

Select one:
A. Image receptor unsharpness
B. Distortion
C. Geometric unsharpness
D. Motion unsharpness
D. Motion unsharpness
What affect does resolution have on recorded detail?

Select one:
A. no affect
B. inverse
C. indirect
D. direct
D. direct
SID is important as a photographic property and a geometric property.

Select one:
True
False
True
Based on the following factors, what is the geometric unsharpness? (In mm)
FSS=1.2mm, SID=54 inches, OID=4 inches
0.09 mm
A misalignment the central ray, radiographed part, or image receptor will cause distortion.

Select one:
True
False
True
Shape distortion can be useful in radiographic imaging.

Select one:
True
False

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True
Which is better at contrast resolution?

Select one:
a. digital
b. film
a. digital
Increasing OID _____ recorded detail.

Select one:
A. has no effect on
B. increases
C. opens a can of
D. decreases
D. decreases
Which combination of factors results in the optimal image?

Select one:
A. max recorded detail and min distortion
B. min recorded detail and max distortion
C. min recorded detail and min distortion
D. max recorded detail and max distortion
A. max recorded detail and min distortion
Shape distortion includes:

Select one:
A. magnification
B. size distortion
C. elongation
d. all of the above
C. elongation
Radiographic misrepresentation of the size or shape of the anatomic structure being imaged is:

Select one:
a. magnification
b. elongation
c. foreshortening
d. all of the above
d. all of the above
Magnification always results in reduced recorded detail.

Select one:
True
False
True
Which of these factors ONLY affects recorded detail?

Select one:
a. Focal spot size
b. Film screens
c. Resolution

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d. SOD
e. SID
f. Patient motion
g. OID
Focal spot size
List 2 types of additive pathologies for each of the abdomen, chest, and skeleton.
Abdomen - aortic aneurysm, calcified stones.
Chest - pneumonia, atelectasis.
Skeleton - metastases, hydrocephalus.
As compared to a patient with a 24 cm thick abdomen, imaging a patient with a 20 cm thick abdomen will result in _____ scatter being produced and _____ image contrast.

Select one:
A. less; lower
B. more; higher
C. less; higher
D. more; lower
C. less; higher
List 2 types of destructive pathologies for each of the abdomen, chest, and skeleton.
Abdomen - aerophagia, bowel obstruction.
Chest - emphysema, pneumothorax.
Skeleton - gout, osteoporosis.

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Air – negative contrast agent
Requires no change in exposure – negative contrast agent
Increases attenuation – positive contrast agent
Barium – positive contrast agent
Require increased exposure – positive contrast agent
Iodine – positive contrast agent
Decreases attenuation – negative contrast agent
With film-screen imaging, for a given exposure technique, increasing part thickness decreases radiographic density.

Select one:
True
False
True
Generally speaking, what is the kVp value for a pediatric (less than 6 years old) skull compared with the adult skull?

Select one:
A. 15% less
B. 5% less
C. 50% more
D. 50% less
A. 15% less
What affect does an increased body part thickness have on scatter?

Select one:
A. increases
B. stays the same
C. decreases
D. no affect
A. increases
A misalignment in which of the following factors will cause distortion? (choose all that apply)

Select one or more:
a. body part
b. image receptor
c. x-ray tube
d. entry or exit point of the central ray
x-ray tube, body part, image receptor, entry or exit point of the central ray
With all other factors remaining the same, a patient with which body habitus would require the highest exposure factors?

Select one:
A. Sthenic
B. Asthenic
C. Hyposthenic
D. Hypersthenic
D. Hypersthenic
Generally speaking, what do additive pathologic conditions require?

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Select one:
A. Increased SID
B. Increased focal spot size
C. Increased mA
D. Increased kVp
D. Increased kVp
Match the following subject contrasts to the correct terms

B – low subject contrast, Chest – high subject contrast, Abdomen – low subject contrast, A – high subject contrast
Which of the following is a destructive pathology?

Select one:
A. Pneumonia
B. Emphysema
C. Pleural effusion
D. Edema
B. Emphysema
With digital imaging, the same mAs and kVp should be used for an AP lumbar spine and a lateral lumbar spine.

Select one:
True
False
False
Assuming all produce appropriate density images, which of the following would be the best exposure technique choice when performing a chest radiograph on an infant?

Select one:
A. Use 200 ms exposure time.
B. Use 60 ms exposure time.
C. Use 40 ms exposure time.

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D. Use at least 80 kVp.
C. Use 40 ms exposure time.
You should always increase your technique when imaging a body part that is stabilized by a splint.

Select one:
True
False
False
What affect does increased scatter have on contrast?

Select one:
A. no affect
B. increases
C. stays the same
D. decreases
D. decreases
Exposure factors used for adult skulls can be used for pediatric patients 6 years old and older.

Select one:
True
False
True
Which one of the following immobilizing devices requires an increase in the exposure technique?

Select one:
A. Fiberglass cast
B. Air splint
C. B and C
D. Plaster cast
D. Plaster cast
In general, what should you do to your technique when imaging a body part with a cast?

Select one:
A. Phone a friend
B. Decrease technique
C. Don't change technique
D. Increase technique
D. Increase technique
In order to image a structure that is located anteriorly in the body, it is best radiographed to minimize magnification by doing a(n) _____ projection.

Select one:
A. Lateral
B. Posterior-anterior
C. Oblique
D. Anterior-posterior
B. Posterior-anterior
What is the purpose of technique charts? (Give at least 2 reasons)
The purposes of technique charts are: to help produce consistent images, satisfy ALARA, and to expand the life of the tube.
What is the appropriate setting for backup time/mAs?

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Select one:
A. 50% of the expected mAs
B. 150% of the expected mAs
C. 400% of the expected mAs
D. 100% of the expected mAs
B. 150% of the expected mAs
A variable kVp-fixed mAs chart may be most effective with:

Select one:
A. Patients imaged with mobile equipment
B. Geriatric patients
C. Trauma patients
D. Pediatric patients
D. Pediatric patients
When the radiographer is considering whether to repeat an image using AEC, it is important to determine the reason the first radiograph failed.

Select one:
True
False
True
AEC devices directly control the:

Select one:
A. SID and OID
B. mA
C. exposure time
D. kVp
C. exposure time
What is the purpose of setting a backup time/mAs?

Select one:
A. To prevent the selection of the wrong detector
B. To make certain the density is acceptable if the patient moves
C. To prevent excessive exposure of the patient
D. To make certain the contrast is acceptable if the chosen kVp is too low
C. To prevent excessive exposure of the patient
Exposure time will _____ using AEC when patient thickness decreases.

Select one:
A. double
B. increase
C. remain the same
D. decrease
D. decrease
Why was automatic exposure control (AEC) originally developed?

Select one:
A. To increase productivity and patient dose
B. To reduce repeats and improve image quality

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C. To decrease the need for technique charts
D. To make the radiographer more of a "button pushing monkey"
B. To reduce repeats and improve image quality
List and BRIEFLY discuss 4 technical considerations when using AEC.
Four technical considerations when using AEC are: proper centering of the part, correct detector selection, kVp and mA selections, and your density selection.
Why are calipers used?

Select one:
A. Adjust for variable grid factors
B. Calculate the proper SID
C. Hold the IR
D. Measure body part thickness
D. Measure body part thickness
What would the result be if a film-screen radiographic examination required an exposure time shorter than the minimum response time of the AEC sensors?

Select one:
A. Image density would be excessive.
B. Image density would be optimal.
C. Response time has nothing to do with AEC.
D. Image contrast would be high.
A. Image density would be excessive.
Select the correct 2 types of technique charts.

Select one or more:
a. Variable kVp/variable mAs
b. Variable kVp/fixed mAs
c. Fixed kVp/fixed mAs
d. Fixed kVp/variable mAs
Variable kVp/fixed mAs, Fixed kVp/variable mAs
Technique charts should follow which principle?

Select one:
A. ALARA
B. JRCERT
C. ASIS
D. HARMS
E. BANANARAMA
A. ALARA
In order to become familiar with manual technical factors (i.e., setting mAs and kVp and not using AEC), it is very helpful to pay attention to the:

Select one:
A. Exposure time
B. mAs readout
C. Heat units
D. Density control
B. mAs readout
It is correct to call all AEC devices phototimers.

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Select one:
True
False
False
If a part measures 14 cm and requires the use of 75 kVp, how much kVp would a part measuring 18 cm require when using a variable kVp-fixed mAs technique chart?

Select one:
A. 83 kVp
B. 67 kVp
C. 91 kVp
D. 77 kVp
A. 83 kVp
When using a fixed kVp-variable mAs technique chart, if part thickness increases by 5 cm, what needs to happen to the mAs?

Select one:
A. It should be tripled.
B. It should be doubled.
C. It should be quartered.
D. It should be halved.
B. It should be doubled.
Technique charts help in which of the following areas? (Choose all that apply)

Select one or more:
a. Provide uncertainty in radiographic factors
b. Reduce exposure
c. Reduce repeats
d. Increase patient dose
e. Produce consistency in image quality
Produce consistency in image quality, Reduce repeats, Reduce exposure
Where is the sensor located in an automatic exposure device that converts light into an electrical signal?

Select one:
A. In front of the patient
B. Behind the image receptor
C. On the tabletop
D. In front of the image receptor
B. Behind the image receptor
Match the following descriptions to the proper type of AEC device.

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More sophisticated – Phototimers
Less prone to failure – Ionization chambers
More accurate – Phototimers
Most common AEC device – Ionization chambers
Which automatic exposure device works by converting x-ray photons first into light and then into an electronic signal?

Select one:
A. Gas detector
B. Backup timer
C. Ionization chamber
D. Phototimer
D. Phototimer
When using AEC with film-screen, what should the radiographer adjust to manipulate the overall density of the image?

Select one:
A. Backup time
B. Density control
C. Focal spot size
D. kVp
B. Density control
With digital imaging, AEC can be used with tabletop exams.

Select one:
True
False
False
Where is the sensor located in an automatic exposure device with a gas-filled chamber?

Select one:
A. On the tabletop
B. In front of the patient
C. Behind the Bucky tray
D. In front of the image receptor
D. In front of the image receptor
What is the appropriate change in kVp when using a variable kVp-fixed mAs technique chart for a 1 cm change in tissue thickness?

Select one:
A. 4 kVp
B. 2 kVp
C. 10 kVp

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D. 6 kVp
B. 2 kVp
What kind of chart uses a kVp value that is high enough to adequately penetrate the part but does not diminish radiographic contrast?

Select one:
A. A and C
B. A variable kVp technique chart
C. A fixed mAs technique chart
D. A fixed kVp technique chart
D. A fixed kVp technique chart
When comparing how radiation exposure is converted into electrical energy, what is the difference between phototimers and ionization chambers?
Phototimers make light with photons which converts into electrical signals. Ionization chambers are cells containing air that interacts with photons to create an electrical charge.
Technique charts are still needed when using AEC.

Select one:
True
False
True
Briefly list and explain the 5 steps in technique chart development.
Step 1 - Determine phantom and radiographic factors
Step 2 - kVp and mAs using 15% rule
Step 3 - Eliminate unacceptable radiographs
Step 4 - Choose optimal radiograph
Step 5 - Extrapolate chart based on optimal image
Each body part requires these steps. Pick your part and make multiple images with varying techniques and choose the optimal image/technique. Use that as you technique on the chart.
AEC devices help control:

Select one:
A. density
B. distortion
C. contrast
D. recorded detail
A. density
A technique chart should be established for:

Select one:
A. Each radiographer
B. Each radiographic tube
C. A and C
D. Each radiologist
B. Each radiographic tube
A good time to use AEC is when the radiographer is unable to accurately center the x-ray beam

Select one:
True
False
False

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Accurate patient measurement is most critical for the:

Select one:
A. Variable kVp-variable mAs technique chart design
B. Fixed kVp-fixed mAs technique chart design
C. Variable kVp-fixed mAs technique chart design
D. Fixed kVp-variable mAs technique chart design
C. Variable kVp-fixed mAs technique chart design
What describes the shortest exposure time required for the AEC device to operate?

Select one:
A. A and C
B. Master density time
C. Minimum response time
D. Backup time
C. Minimum response time
When doing a CR study using AEC, what will happen to the patient exposure when changing from the -1 to +1 density setting?

Select one:
A. Patient exposure will decrease
B. Patient exposure will increase
C. Patient exposure will not change
B. Patient exposure will increase
When exposure factors are selected using APR, the variables should not be adjusted.

Select one:
True
False
False
Which automatic exposure device works by using a gas-filled chamber?

Select one:
A. Ionization chamber
B. Diode sensor
C. Bucky proliferator
D. Phototimer
A. Ionization chamber
Which factors on a technique chart are standardized? (Choose all that apply)

Select one or more:
a. SID
b. Measuring point
c. IR
d. Grid ratio
e. mAs
f. FSS
g. kVp
Measuring point
SID
IR

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Grid ratio
FSS
When doing a film-screen study using AEC, what will happen to the density in the area of interest when changing from 70 kVp to 90 kVp?

Select one:
A. Density will increase
B. Density will not change
C. Density will decrease

B. Density will not change

Term
The relationship between screen speed and resolution is:
Select one:
a. inverse
b. direct
Definition
a. inverse
Term
The primary controlling factor of density is
Select one:
a. focal spot
b. mAs
c. kVp
d. SID
e. filtration
Definition
b. mAs
Term
Any irregularity on an image that is not caused by the proper shadowing of tissue by the primary beam
Select one:
a. artifact
b. elongation
c. foreshortening
d. distortion
Definition
d. distortion
Term
Which of the following sets of factors will produce the x-ray image with the shortest scale of contrast?

mA time kVp grid

a. 200 .02 85 6:1

b. 300 .10 65 16:1

c. 400 .05 70 8:1

d. 200 .15 75 12:1
Select one:

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a. a
b. c
c. d
d. b
Definition
d. b
Term
In order to notice a visible change in a film's density, by how much must the mAs be increased or decreased?
Select one:
a. 30%
b. 15%
c. 20%
d. you must always half or double your mAs
Definition
a. 30%
Term
Patient motion will result in a decrease of what on a radiograph?

1. resolution

2. contrast

3. density
Select one:
a. 1 only
b. 2 only
c. 1 and 2
d. 2 and 3
e. 1, 2, and 3
Definition
a. 1 only
Term
Which of the following will result in the least amount of patient exposure?
Select one:
a. increase kVp, increase mAs
b. decrease kVp, increase mAs
c. decrease kVp, decrease mAs
d. increase kVp, decrease mAs
Definition
d. increase kVp, decrease mAs
Term
A radiograph is produced with too little contrast and excessive radiographic density. Which of the following changes will best correct this error?
Select one:
a. increase the mAs, increase the kVp
b. decrease the mAs, increase the kVp
c. increase the mAs, decrease the kVp
d. decrease the mAs, decrease the kVp
Definition
d. decrease the mAs, decrease the kVp
Term

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Using the small filament in the x-ray tube versus using the large filament will result in a:

1. increase in contrast

2. decrease in distortion

3. decrease in density

4. increase in resolution

Select one:
a. 3 only
b. 4 only
c. 1 and 2
d. 2 and 3
e. 1 and 4
Definition
b. 4 only
Term
Which of the following sets of technical factors will produce a radiograph with the greatest amount of recorded detail?

mAs kVp grid screen focal ss

a. 20 85 8:1 100 1 mm

b. 40 70 12:1 400 2 mm

c. 35 62 6:1 400 1 mm

d. 60 80 10:1 100 2 mm
Select one:
a. c
b. a
c. b
d. d
Definition
b. a
Term
The overall blackening on a film best describes
Select one:
a. resolution
b. contrast
c. distortion
d. density
Definition
d. density
Term
The amount of magnification on a film is determined by:
Select one:

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a. SOD/SID
b. OID/SOD
c. SID/SOD
d. SOD/OID
Definition
c. SID/SOD
Term
When the central ray is perpendicular to the image receptor, but the part is neither perpendicular to the central ray nor parallel to the image receptor, _____ occurs:
Select one:
a. elongation
b. foreshortening
Definition
a. elongation
Term
Which of the following factors will increase radiographic contrast?

1. using a slower screen speed

2. decreasing collimation

3. using a higher grid ratio
Select one:
a. 1 only
b. 3 only
c. 1 and 2
d. 2 and 3
e. 1, 2, and 3
Definition
b. 3 only
Term
Which of the following sets of technical factors will produce a radiograph with the greatest amount of magnification?

mAs kVp SID OID FSS

a. 15 60 140 cm 20 cm 2 mm

b. 30 80 180 cm 20 cm 1 mm

c. 25 70 180 cm 30 cm .3 mm

d. 50 65 100 cm 30 cm .3 mm
Select one:
a. b
b. c
c. a
d. d
Definition
d. d
Term

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A change in which geometric factor will not effect the magnification of a radiographic image?
Select one:
a. SOD
b. OID
c. focal spot
d. SID
Definition
c. focal spot
Term
Recorded detail can be improved by:
Select one:
a. collimation
b. decreasing the SID
c. decreasing the kVp
d. decreasing the phosphor crystal size
Definition
d. decreasing the phosphor crystal size
Term
The utilization of 120 kVp will result in:
Select one:
a. high exposure latitude and short scale contrast
b. low exposure latitude and short scale contrast
c. high exposure latitude and long scale contrast
d. low exposure latitude and long scale contrast
Definition
c. high exposure latitude and long scale contrast
Term
Which of the following has no impact on contrast?

1. kVp

2. focal spot size

3. filtration

4. collimation
Select one:
a. 1 only
b. 2 only
c. 2 and 3
d. 1, 3, and 4
e. 1, 2, 3, and 4
Definition
b. 2 only
Term
This type of artifact results in emulsion pickoff, gelatin buildup- sludge and can appear as sharp positive or negative optical density on film
Select one:
a. roller mark
b. dirty rollers
c. chemical fog

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d. wet pressure sensitization
Definition
b. dirty rollers
Term
Which of the following factors will affect the visibility of detail?

1. OID

2. SID

3. mAs

4. focal spot size
Select one:
a. 1, 2, and 4
b. 3 only
c. 3 and 4
d. 2 and 4
e. 1 and 2
Definition
b. 3 only
Term
An increase in radiographic density can result from an increase in:

1. SID

2. kVp

3. mAs
Select one:
a. 1 and 2
b. 1 and 3
c. 2 and 3
d. 1, 2, and 3
Definition
c. 2 and 3
Term
The penetrability of an x-ray beam is determined by
Select one:
a. kVp
b. mAs
c. grid ratio
d. screen speed
Definition
a. kVp
Term
How much a technologist can change or vary an exposure technique without compromising image quality is referred to as:
Select one:
a. latitude
b. contrast

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c. quantum mottle
d. resolution
Definition
a. latitude
Term
Recorded detail can be improved by doing the following:

1. increasing the OID

2. increasing the SID

3. decreasing the OID

4. decreasing the focal spot size
Select one:
a. 1 and 2
b. 2 and 3
c. 1, 3, and 4
d. 2, 3, and 4
Definition
d. 2, 3, and 4
Term
What effect does changing the kVp from 70 to 55 have on an image?
Select one:
a. decreases contrast
b. increases contrast
c. produces a low contrast image
d. produces a long scale of contrast
Definition
b. increases contrast
Term
A reduction in scatter radiation is likely to occur from which of the following:

1. increased kVp

2. decreased collimation

3. increase in grid ratio

4. decrease in tissue volume
Select one:
a. 3 only
b. 1 and 3
c. 2 and 3
d. 3 and 4
e. 1, 2, and 4
Definition
d. 3 and 4
Term
Which of the following kVp values will produce an x-ray with the shortest scale of contrast?
Select one:

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a. 90
b. 75
c. 62
d. 50
Definition
d. 50
Term
Adding filtration to an x-ray beam will
Select one:
a. increase the average energy of the beam
b. increase the recorded detail
c. increase the contrast
d. increase the density
Definition
a. increase the average energy of the beam
Term
The primary controlling factor of contrast is
Select one:
a. kVp
b. distance
c. mAs
d. filtration
Definition
a. kVp
Term
Artifacts from radiation and light fog can be caused by —heat below 68 degrees and 40 to 60 % humidity.
Select one:
True
False
Definition
True
Term
Which of the following sets of technical factors will result in a film with the least amount of chance for motion?
Select one:
a. 500 mA, .04 sec, 80 kVp
b. 400 mA, .05 sec, 60 kVp
c. none of the above, they all have the same mAs
d. 100 mA, .2 sec, 65 kVp
e. 200 mA, .1 sec, 75 kVp
Definition
a. 500 mA, .04 sec, 80 kVp
Term
Low kVp produces which of the following?

1. high contrast

2. few gray tones

3. long scale contrast

4. short scale contrast

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5. low contrast

6. many gray tones
Select one:
a. 1 only
b. 5 only
c. 3 and 5
d. 1, 2, and 4
e. 3, 5, and 6
Definition
d. 1, 2, and 4
Term
The purpose of mAs is to
Select one:
a. regulate density
b. provide higher contrast
c. increase recorded detail
d. increase x-ray beam penetration
Definition
a. regulate density
Term
Radiographic contrast can best be described as:
Select one:
a. the overall blackening on a radiograph
b. the difference between densities
c. how black and white an image appears
d. the amount of grayness on a radiograph
Definition
b. the difference between densities
Term
Magnification is another name for:
Select one:
a. resolution
b. penumbra
c. shape distortion
d. size distortion
Definition
d. size distortion
Term
Electron build up on the emulsion due to low humidity is referred to as static
Select one:
True Correct
False
Definition
True
Term
A film is said to possess excessive radiographic density. This would appear:
Select one:
a. to have poor recorded detail
b. to have excessive contrast

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c. darker than normal
d. lighter than normal
Definition
c. darker than normal
Term
A technique chart calls for 10 mAs at 72 kVp for a particular x-ray. An unacceptable image is produced using 10 mAs at 62 kVp; the image is too light. How can this problem best be corrected?
Select one:
a. increase the mAs
b. decrease the kVp
c. decrease the mAs
d. increase the kVp
e. decrease the SID
Definition
d. increase the kVp
Term
Given a baseline technique of 200 mA, .1 sec, 70 kVp, 40" SID, 12:1 grid, 400 screen, indicate what effect the following changes will have on contrast:

- mark "A" for increase

- mark "B" for decrease

- mark "C" for no change

80 kVp

100 screen

300 mA .2 sec

16:1 grid
Definition
80 kVp
– b,

100 screen
– c,

300 mA .2 sec
– c,

16:1 grid
– a
Term
The white areas on a radiograph are the result of what type of attenuation?
Select one:
a. scatter
b. absorption
Definition
b. absorption

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Term
Beam restriction has the following effect on contrast
Select one:
a. decreases the contrast by decreasing the field size
b. increases the contrast by reducing the volume of tissue irradiated
c. increases the contrast by increasing the field size
d. decreases the contrast by reducing the volume of tissue irradiated
Definition
b. increases the contrast by reducing the volume of tissue irradiated
Term
Improper tube, object and image receptor alignment will result in:
Select one:
a. shape distortion
b. motion distortion
c. size distortion
d. lack of recorded detail
Definition
a. shape distortion
Term
An AP abdomen is taken using 75 kVp, 40 mAs, 40" SID, 400 screen, 12:1 grid, 1.2 mm focal spot. The kidneys lie 4" from the back of the patient's body. In addition, the patient's back is 3" from the bucky. What is the penumbra (focal spot blur) of the kidneys?
Select one:
a. .1 mm
b. .129mm Incorrect
c. .097 mm
d. .25 mm
Definition
d. .25 mm
Term
The first method technologists should use to control voluntary motion:
Select one:
a. restraining devices
b. sandbags
c. adequate communication
Definition
c. adequate communication
Term
An AP abdomen is taken using 75 kVp, 40 mAs, 40" SID, 400 screen, 12:1 grid, 1.2 mm focal spot. The kidneys lie 4" from the back of the patient's body. In addition, the patient's back is 3" from the bucky. What is the magnification factor of the kidneys?
Select one:
a. 1.08
b. .825
c. 1.21
d. 1.11
Definition
c. 1.21
Term
Given a baseline technique of 300 mA, .5 sec, 65 kVp, 48" SID, 10:1 grid, 200 screen, indicate what effect the following changes will have on density:

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- mark “A” for increase

- mark “B” for decrease

- mark “C” for no change

72" SID

100 screen

90 kVp

200 mA, .8 secs
Definition
72" SID
– b,

100 screen
– b,

90 kVp
– a,

200 mA, .8 secs
– a
Term
An AP abdomen is taken using 75 kVp, 40 mAs, 40” SID, 400 screen, 12:1 grid, 1.2 mm focal spot. The kidneys lie 4” from the back of the patient’s body. In addition, the patient’s back is 3” from the bucky. determine the size of the kidneys on the radiograph. The actual size of the kidneys is 4.1” X 6.2”.
Select one:
a. 4.4" x 6.7"
b. 3.4" x 5.1" Incorrect
c. 4.6" x 6.9"
d. 5.0" x 7.5"
Definition
d. 5.0" x 7.5"
Term
A lack of photons striking an image receptor that results from very fast screens used with low mAs settings causes:
Select one:
a. line pairs per millimeter
b. focal spot blur
c. quantum mottle
d. distortion
Definition
c. quantum mottle
Term
As the SID is increased, the density on the film
Select one:
a. decreases

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b. remains the same
c. increases
Definition
a. decreases
Term
Handling artifacts can only occur prior to processing the film.
Select one:
True
False
Definition
False
Term
Given a baseline technique of 400 mA, .02 sec, 62 kVp, 48” SID, x-ray performed tabletop, tube angled 30 degrees caudal, indicate what effect the following changes will have on distortion (size and shape):

- mark “A” for increase

- mark “B” for decrease

- mark “C” for no change

50 kVp

tube perpendicular to the image receptor

x-ray performed in the bucky

20 mAs
Definition
50 kVp
– c,

tube perpendicular to the image receptor
– b,

x-ray performed in the bucky
– a,

20 mAs
– c
Term
Given a baseline technique of 300 mA, .1 sec, 75 kVp, 60” SID, 3” OID, 12:1 bucky grid, 400 screen, indicate what effect the following changes will have on recorded detail:

- mark “A” for increase

- mark “B” for decrease

- mark “C” for no change

x-ray performed tabletop

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40" SID

70 kVp

100 screen
Definition
x-ray performed tabletop
– a,

40" SID
– b,

70 kVp
– c,

100 screen
– a
Term
Calculate the penumbra (focal spot blur) using a .5 mm focal spot, 40” SID, 5” OID
Select one:
a. .07mm
b. 3.5 mm
c. .06 mm
d. 4 mm
Definition
a. .07mm
Term
An image that is predominately black and white is said to exhibit this type of contrast:

1. short scale

2. low

3. long scale

4. high
Select one:
a. 1 only
b. 2 and 4
c. 2 and 3
d. 1 and 4
e. 1 and 2
Definition
d. 1 and 4

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Which of the following is an assembly that consists of one master roller, several planetary rollers, and two guide shoes?
a.) A crossover rack
b.) A detector assembly
c.) A roller subassembly
d.) A transport rack
e.) A turnaround assembly
Tap card to see definition
e.) A turnaround assembly

2/18
Replenishment of which tank in the processing system is most important?
a.) Developer
b.) Dryer
c.) Fixer
d.) Wash
e.) Wetter
a.) Developer

3/18
When a film is inserted into an automatic processor:
a.) A microswitch grips it.
b.) Guide shoes grip it.
c.) It should be centered on the feed tray.
d.) The long dimension should be against the rail.
e.) The short dimension should be against the rail.
e.) The short dimension should be against the rail.

4/18
Adequate drying is necessary to:
a.) Complete development
b.) Complete fixation
c.) Obtain adequate contrast
d.) Reduce artifacts
e.) Strengthen the base
d.) Reduce artifacts

Between the fixing tank and the wash tank, the film passes through a:
a.) Crossover rack
b.) Drying chamber
c.) Receiving bin
d.) Transportation rack
e.) Turnaround rack
a.) Crossover rack

7/18
Which of the following is not one of the major systems of an automatic processor?

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a.) The development system
b.) The drying system
c.) The electrical system
d.) The replenishment system
e.) The transport system
a.) The development system

8/18
The principal purpose of the circulation system is to:
a.) Agitate the chemistry
b.) Agitate the film
c.) Control chemistry concentration
d.) Control chemistry temperature
e.) Replenish the chemistry
a.) Agitate the chemistry

9/18
Which segment of the circulation system is most important to archival quality?
a.) Developing
b.) Drying
c.) Fixation
d.) Washing
e.) Wetting
c.) Fixation

10/18
Underreplenishment of the developer will result in:
a.) A decrease in contrast
b.) A reduction in spatial resolution
c.) An improvement in spatial resolution
d.) An increase in contrast
e.) No change in contrast
a.) A decrease in contrast

11/18
Replenishment tanks should have close-fitting floating lids primarily:
a.) For control of replenishment rate
b.) For control of temperature
c.) To control aerial oxidation
d.) To easily monitor fluid level
e.) To reduce splash hazard
c.) To control aerial oxidation

12/18
Dry-to-drop time refers to:
a.) Development time
b.) Exposure-to-viewbox time
c.) Passbox-to-receiving bin time
d.) Time from feed tray to receiving bin
e.) Time in the passbox

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d.) Time from feed tray to receiving bin

13/18
The segment of the circulation system that is most likely to contain a filter is the:
a.) Developer
b.) Dryer
c.) Fixer
d.) Wash cycle
e.) Wetting cycle
a.) Developer

14/18
If the power of the drive motor is transferred through a chain, the connecting device is usually a:
a.) Gear
b.) Pulley
c.) Roller
d.) Spring
e.) Sprocket
e.) Sprocket

15/18
Control of the replenishment system is accomplished by a:
a.) Drive motor
b.) Float valve
c.) Microswitch
d.) Timer
e.) Transport rack
c.) Microswitch

17/18
The transport system includes all except which of the following?
a.) Drive motor
b.) Drying chamber
c.) Guide shoes
d.) Rollers
e.) Transport racks
b.) Drying chamber

18/18
The 3-inch master roller is a part of a:
a.) Guide shoe
b.) Microswitch
c.) Planetary roller
d.) Replenishment system
e.) Turnaround assembly
e.) Turnaround assembly

1/10
D

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What term is defined as a measurement of the amount of light transmitted through the film? (p 233)
A. Sensitometry
B. Film Contrast
C.Film Speed
D. Optical Density

2/10
C
What is the diagnostic range of optical densities? (p 235)
A. 0.15 to 4.0
B. 0.5 to 1.25
C. 0.50 to 2.0
D. 0.10 to 2.0

3/10
D
An optical density of 1.0 indicates that _____ light was transmitted. (p 234)
A. 0.01%
B. 0.1%
C. 1.0%
D. 10%

4/10
B
A _____ change in optical density results from a change in the percentage of light transmittance by a factor of 2. (p 235)
A. 0.03
B. 0.3
C. 3.0
D. 30

5/10
A
Changes in exposure have little effect on density in which of the following regions of the sensitometric curve: (1) toe, (2) shoulder, or (3) straight line? (p 239)
A. 1 and 2 only
B. 1 and 3 only
C. 2 and 3 only
D. 1,2, and 3

6/10
C
When the exposure technique used produces densities outside the straight-line portion of a sensitometric curve, how is contrast affected? (p 250)
A. Increased
B. No effect
C. Decreased
D. Improved

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7/10
A

Compare the sensitometric curves in Figure 9-20. Which film is faster?
A. Film A
B. Film B
C. Film C
D. Film D

8/10
D

Compare the sensitometric curves in Figure 9-20. Which film has lower contrast?
A. Film A
B. Film B
C. Film C
D. Film D

9/10
A
Film B has an exposure speed point of 1.5, and Film A has an exposure speed point of 1.2. Is there a difference in speed?
A. Film A is twice as fast as Film B
B. Film A and Film B have equal speed
C. Film A is half as fast as Film B
D. Film B is twice as fast as Film A

10/10
C
Changes in radiation exposure have the greatest effect on optical densities in which sensitometric region?
A. Toe
B. Shoulder
C. Straight line
D. Toe and Shoulder

1/10
C
What is defined as preestablished guidelines used to select standardized exposure factors? (p 256)
A. Comparative anatomy
B. Extrapolation technique
C. Exposure technique chart
D. Automatic exposure control

2/10
B
_____ is the primary patient factor that determines the selection of exposure factors. (p 256)

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A. Age
B. Part measurement
C. Physical condition
D. Weight

3/10
C
A primary goal of exposure technique charts is to: (p 256)
A. extend the life of the x-ray tube
B. improve the radiographer's accuracy
C. consistently produce quality images
D. increase the patient work flow

4/10
A
Which of the following is an important condition required for technique charts to be effective: (p 258)
A. Equipment must be calibrated to perform properly
B. One technique chart should be used for all radiographic units
C. All technologists should use the same mA setting
D. The chart should not be revised once it has been used

5/10
D
Which of the following factors would not be standardized on technique charts: (p 259)
A. Image receptor speed
B. Grid ration
C. SID
D. Patient age

6/10
B
What type of exposure technique system used a fixed mAs regardless of part thickness?
A. Fixed kVp
B. Variable kVp
C. Manual
D. AEC

7/10
C
Of the following, which is most important when using a technique chart?(p 266)
A. One radiographer revises the chart
B. A high mA value is set
C. The part is measured accurately
D. Patient history is included

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8/10
B
What is an advantage of the fixed kVp technique chart? (p 260)
A. It produces higher-contrast images
B. It reduces patient exposure
C. kVp changes are easy to make
D. Smaller technique changes are possible

9/10
D
What is a disadvantage of the variable kVp technique chart? (p 260)
A. It produces lower-contrast images
B. It is difficult to construct
C. It may not be effective with small extremities
D. It increases heat load on the x-ray tube

10/10
A
In creating either type of exposure technique chart, what is most important? (p 263)
A. Achieving adequate penetration of the anatomic part
B. Selecting the same milliamperage value
C. Not requiring radiographers to consult them
D. Producing images with similar radiographic contrast

The "halo artifact" is associated with:
a) Equalization
b) Shuttering
c) Edge enhancement
d) Smoothing
c) Edge enhancement
A bit depth of one (1) results in a display of how many colors?
a) 256
b) 1
c) 2
d) 8
c) 2
1/63

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Terms in this set (63)
The "halo artifact" is associated with:
a) Equalization
b) Shuttering
c) Edge enhancement
d) Smoothing

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c) Edge enhancement
A bit depth of one (1) results in a display of how many colors?
a) 256
b) 1
c) 2
d) 8
c) 2
In a CRT video display, the images are "drawn" on the screen. The speed at which the images are drawn is called the:
a) Refresh rate
b) Detective quantum efficiency
c) Nyquist frequency
d) Modulation transfer function
a) Refresh rate
The modality work list is compiled from information sent to the acquisition workstation from the:
1) HIS
2) PACS
3) RIS
a) 1 and 3
b) 2 and 3
c) 1 and 2
d) 1, 2 and 3
a) 1 and 3
"Metal mottle" is associated with:
a) Shuttering
b) Equalization
c) Smoothing
d) Edge enhancement
b) Equalization
The proper way to correct an image's appearance when a default menu consistently fails to display the image properly is to:
a) Adjust the image brightness and contrast and transmit it to the PACS.
b) Redefine the menu's processing codes to properly display the image.
c) Select a menu that makes the image appear satisfactory.
d) Select a menu from the same anatomic region to preserve spatial resolution.
b) Redefine the menu's processing codes to properly display the image
The personnel who have deep level software privileges particularly associated with screen configurations and data displays are:
a) Physician users
b) Super users
c) All users
d) Technologist users
b) Super users
The delayed light emission from a video display that occurs after an image is refreshed is called:
a) Fluorescence
b) Luminosity
c) Phosphorescence
d) Persistence
d) Persistence
In digital radiography, image brightness on a CRT is adjusted by the/a:
a) Window width

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b) Scanning lines
c) kVp increase
d) Window level
d) Window level
In digital radiography, the radiographic contrast on a CRT can be adjusted by changing the:
a) Window level
b) mAs
c) Matrix size
d) Window width
d) Window width
If the goal is to achieve better spatial resolution from a video display, then dot pitch:
a) Changes with bandwidth
b) Has no effect
c) Must increase
d) Must decrease
c) Must increase
The current ACR standards for a gray-scale video display's minimum luminance is:
a) 800 candelas/meter squared
b) 1,500 candelas/meter squared
c) 3,000 candelas/meter squared
d) 160 candelas/meter squared
d) 160 mandelas/meter squared
The ratio of the video display screen's width to its height defines the:
a) Vertical ratio
b) Contrast ratio
c) Display ratio
d) Aspect ratio
d) Aspect ratio
Altering which of the following image characteristics may result in permanent data loss from an image data set?
1) Image brightness
2) Image contrast
3) ROI location
a) 1 and 3
b) 1, 2 and 3
c) 1 and 2
d) 2 and 3
b) 1, 2, and 3
The difference between maximum and minimum luminance of a video display is known as:
a) Black level
b) Contrast resolution
c) Horizontal resolution
d) Vertical resolution
b) Contrast resolution

15/15
A
A common plus-density artifact caused from bending the film is: (p 223)
A. half-moon marks
B. static discharge
C. abrasion

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D. fogging

14/15
B
The type of silver-recovery unit that uses and electrically charged drum to attract the silver is called a(n) _____ unit: (p 222)
A. an extraction filter
B. electrolytic
C. steel wool
D. metallic replacement

13/15
C
Safelight filters are chosen based on the: (p 220)
A. amount of light intensity
B. dimensions of the darkroom
C. film sensitivity
D. power rating

12/15
D
Under what environmental conditions should radiographic film be stored (p 218)
A. Temperature between 40 deg and 60 deg F and relative humidity between 4% and 60%
B. Temperature between 50 deg and 70 deg F and relative humidity between 50% and 70%
C. Temperature between 40 deg and 60 deg F and relative humidity between 50% and 70%
D. Temperature between 50 deg and 70 deg F and relative humidity between 40% and 60%

11/15
C
Decreasing the developer temperature (p 215)
A. decreases oxidation
B. increases contrast
C. decreases density
D. increases processing time

10/15
A
Processing chemicals must be replenished to maintain activity and volume when depleted primarily by: (p 213)
A. exhaustion
B. diffusion
C. precipitation
D. condensation

9/15
D
The type of roller responsible for moving the film from one tank to another is a(n): (p 209)
A. transport
B. entrance
C. turnaround
D. crossover

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8/15
C
The type of roller responsible for moving the film from the bottom of the tank upward is a(n) _____ roller: (p 209)
A. transport
B. entrance
C. turnaround
D. crossover

7/15
B
Finished radiographs should retain what percentage of their moisture? (p 226)
A. 2% - 5%
B. 10% - 15%
C. 20% - 30%
D. 35% - 45%

6/15
C
Staining or fading of the permanent image results when too much _____ remains on the film with proper washing. (p 206)
A. phenidone
B. acetic acid
C. thiosulfate
D. glutaraldehyde

5/15
D
The fixing agent used to clear the undeveloped silver halide crystals is: (p 226)
A. hydroquinone
B. aluminum chloride
C. potassium bromide
D. ammonium thiosulfate

4/15
A
The pH environment required for proper development of exposed film is: (p 205)
A. alkaline
B. oxidized
C. acetic
D. exhausted

3/15
C
Which of the following solutions are responsible for reducing the exposed silver halide crystals to black metallic silver: (1) ammonium thiosulfate (2) hydroquinone or (3) phenidone (p 205)
A. 1 and 2 only
B. 1 and 3 only
C. 2 and 3 only

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D. 1,2, and 3

2/15
C
The sequential order for processing radiographic film is: (p 204-207)
A. developing, washing, fixing, dryer
B. fixing, washing, developing, dryer
C. developing, fixing, washing,dryer
D. Fixing, developing, washing, dryer

1/15
B
Conversion of the latent image to the manifest image is accomplished by:(p 202)
A. the Gurney-Mott theory
B. Radiographic processing
C. diffusion
D. oxidation
Tap card to see term

Answer:
The correct answer for the question "Current diagnosis of idiopathic pulmonary haemosiderosis (IPH) is confirmed by which one of the following?:" is:

3. Bronchoalveolar lavage (BAL).

The correct answer for the question "The causes of postpartum haemorrhage include all of the following except" is:

4. Massaging the fundus of uterus after placental delivery

Answer:
The correct answer for the question "Sarcoidosis affects the kidneys most commonly by:" is:

2. Nephrocalcinosis secondary to overproduction of calcitriol from extrarenal granulomata
"Patients with vascular Ehlers-Danlos Syndrome will usually present to the Emergency Room with?" is:

3. Vascular hemorrhages and gastrointestinal bleeds.
Answer:
The correct answer for the question "What is the histological hallmark of sarcoidosis?" is:

1. Non-caseating epithelioid cell granulomas.
The correct answers for the question "Following what process(es) does a plica become symptomatic?" are:

1. Idiopathic

2. Trauma

3. Inflammation

4. Rupture

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The correct answer for the question "Which of the following is true of abandoned VP shunt migration and bowel perforation?" is:

4. Abandoned VP shunt migration and bowel perforation can be a chronic erosive process which develops slowly through local erosion.
The correct answer for the question "What are the CTA findings of coronary sinus communication to the left atrium?" is:

1. a bridging vein connecting the coronary sinus to the left atrium

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